



A recommender-based web platform to boost tourism in marginal territories

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Abstract

Recommender engines are software applications employed in online tourism information searches to suggest useful content and guide user choices. They represent a significant area of research within the tourism sector, crucially influencing traveler decisions through personalized recommendations. This paper focuses specifically on the Branding4Resilience (B4R) project, whose objective is to promote inner areas, by co-designing virtuous transformations with their inhabitants through novel branding strategies and digital enabling infrastructures. All experiments and assessments described in this study were conducted within the Italian area of Sassoferrato, corresponding to the criteria defined by the B4R initiative. The paper introduces a dedicated web platform aimed at enhancing tourism by directing attention towards inland destinations and networking opportunities. The platform enables local providers of tourism-related services, such as guides, event organizers, and restaurateurs, to connect effectively with tourists seeking distinctive experiences. By facilitating interactions, the platform promotes collaboration, innovation, and the co-creation of personalized experiences, transforming tourists into active participants and contributors. To evaluate user experience, perceived usefulness, and innovation of the platform, we analyzed user responses to a structured questionnaire, which combined frequently addressed questions from existing literature with newly developed items specific to our platform. Constructs included tourist platform usage, perceived utility, content quality, design and usability, and innovative elements. The implemented algorithm demonstrated an average sensitivity of 79.74%, highlighting its efficacy in providing relevant suggestions to the platform's “consumers” users.

Keywords Recommender systems · Branding · Machine learning for tourism · Collaborative web platform

1 Introduction

Tourism is a crucial sector for the Italian economy, representing a significant source of income and employment. However, the benefits of tourism are not evenly distributed across the country, and many marginal territories struggle to attract visitors and compete with more established destinations. In recent years, new technologies have emerged as a potential solution to this problem, offering innovative ways to promote and enhance tourism, particularly in underdeveloped areas (Gavalas and Kenteris 2011) (Ricci et al. 2015). The integration of web platforms and recommender engines has emerged as a key strategy for promoting lesser-known areas and stimulating economic development. Using digital tools, regions can increase their visibility, attract tourism, and support local economies.

Building on these opportunities, the Branding4Resilience (B4R) (Branding4Resilience 2024) project was launched as a four-year research (2020–24) and design initiative of national interest funded by the Italian Ministry of Education, University and Research (MIUR). Its objective is to promote inner areas as fragile and marginal territories characterized by unique cultural and natural resources, by co-designing a virtuous transformation with their inhabitants based on novel branding strategies and digital enabling infrastructures. The project formulates a concept of branding, which is a comprehensive approach to shaping the identity and perception of a territory through the synergy between local communities and actors, and external visitors. The aim is to elevate the quality and resilience of these territories. The B4R project focuses especially on four Italian regions: Piedmont, Trentino, Sicily, and Marche. Branding and Identity Regional branding is a powerful tool that transforms local identities into marketable products, allowing regions to capitalize on their unique cultural, historical, and natural assets. Successful case studies in West Cork, Groene Woud, and Pajottenland demonstrate how effective branding can create a strong regional identity that resonates with both residents and visitors. Digital tools and the advent of web platforms have significantly reduced barriers to international tourism, enhancing the visibility of lesser-known destinations and making it easier for travelers to discover and access these areas (Cillo et al. 2021). Social media, travel websites, and mobile applications, have revolutionized the way regions promote themselves and interact with potential visitors. Taking advantage of these insights, regions can tailor their offerings to meet the needs and expectations of different target audiences, thereby enhancing visitor satisfaction and encouraging repeat visits (Bergemann and Bonatti 2024) (Nwaimo et al. 2024).

In this context, recommender systems play an increasingly important role in regional branding and tourism promotion. Recommender systems are digital tools that use algorithms to provide personalized suggestions to users based on their preferences, behaviors, and historical data. By integrating recommender systems into regional development platforms and tourism websites, regions can offer tailored recommendations for attractions, events, and activities, thereby enhancing the visitor experience and increasing engagement. These systems can help travelers discover hidden gems and lesser-known destinations that align with their interests, contributing to a more personalized and satisfying travel experience (Zhao et al. 2014). Recommender systems also enable regions to optimize their marketing efforts by tar-

getting specific segments of the audience with content that is most likely to resonate with them, thereby increasing the effectiveness of promotional campaigns (Ardissono et al. 2002). However, they also face several challenges, particularly in the context of promoting lesser-known regions. One of the main challenges is the cold start problem, which occurs when there is insufficient data to generate accurate recommendations. This issue is particularly relevant for lesser-known destinations that may lack a substantial amount of user-generated content or visitor data. The cold start problem can hinder the effectiveness of recommender systems in providing personalized suggestions, especially for new users or new attractions that have not yet received sufficient interactions. Addressing this issue is crucial to ensuring that recommender systems can effectively promote lesser-known areas and contribute to the overall success of regional branding efforts (Xie et al. 2020) (Sethi and Mehrotra 2021). Nevertheless, while promoting tourism in lesser-known destinations can alleviate crowding in popular hotspots, it is also essential to recognize that an unregulated surge of visitors to fragile or marginal areas may replicate the same adverse effects associated with mass tourism. The phenomenon commonly referred to as “overtourism”, where tourist numbers exceed an area’s carrying capacity, can precipitate environmental degradation, cultural commodification, and the displacement of local residents (Koens et al. 2018). Excessive visitation pressures can also spark social tensions, drive up living costs, and diminish the overall quality of life for local inhabitants (Peeters et al. 2021). Researchers must thoughtfully calibrate the initiative to bolster economic opportunities without inadvertently spawning a new wave of mass tourism detrimental to both the environment and community well-being.

One of the primary goals of the B4R project is the design and development of the B4R Platform, a collaborative web platform tailored to co-create highly personalized itineraries for visitors and temporary residents exploring the studied inland region. The platform facilitates access to experiences offered by local operators, institutions, and communities. A key aspect of its design is the integration of storytelling techniques and user profiling, which enables the system to collect and connect data from the focus areas. The platform seeks to transcend existing solutions by offering a more engaging and immersive experience, moving beyond conventional information systems that simply match content to users’ preferences. Instead, it empowers users to actively participate in the creation of offerings, positioning them as central figures in the organization of events and the customization of their experiences. This approach transforms tourists into active contributors and ambassadors for the destination (Gretzel et al. 2006). By fostering such engagement, the platform not only enriches the tourism experience but also cultivates a sense of ownership and community among users, strengthening the relationship between visitors and the local territory. Specifically, the platform implements a new approach to territorial branding based on scenario and pattern analysis, allowing for the customization of offerings such as mobility, cultural events, and tourist proposals according to individual user profiles.

Although the project belongs to a broader context, this article focuses on a tourism case study, in particular within the broader field of intelligent tourism recommender systems, whose characteristics will be deepened in the following section, providing a comprehensive and tailored recommendation experience and unresolved questions,

particularly those related to effectively promoting lesser-known destinations. By integrating a hybrid approach, our system aims to tackle issues such as the cold start problem, which is a major hurdle in promoting lesser-known areas.

Active participation enables users to become co-creators of their journeys, adding a layer of personalization and authenticity that goes beyond what traditional systems can offer (De Bruin and Jelinčić 2016). Consequently, the platform aims to elevate the quality and resilience of regions by co-designing transformative strategies for places in collaboration with their local communities, thus enhancing the overall tourism landscape. The team tested the B4R Platform in the Appennino Basso Pesarese-Anconetano region, involving several municipalities.

This paper investigates whether a collaborative web platform can facilitate the encounter between visitor and territory, mainly in inner areas, matching the offer of cultural, tourist and business resources, and guide visitors in co-creating personalized experiential paths. In particular, it presents the platform's design and implementation, highlighting its innovative approach to personalized recommendations and storytelling and the experimental results in terms of achieved platform's user experience, perceived usefulness, and innovative elements.

At the current stage, the following points summarize the main contributions of our research work:

1. Recommender-based collaborative web platform: we designed and implemented a collaborative web platform based on a recommender system engine aimed at enhancing the tourism experience by co-creating highly personalized itineraries. This platform connects visitors with local operators, institutions, and communities, facilitating the discovery of cultural, tourist, and business resources in the inland area analyzed.

2. Hybrid Recommender Algorithm: our recommender algorithm employs a hybrid approach, combining content-based and collaborative filtering with k-Nearest Neighbors (k-NN). This approach is specifically designed to address the cold start problem, which is crucial in cases with limited data, such as ours, and the filter bubble effect. Given that Sassoferrato is a small community with a relatively low number of activities, the scarcity of data makes addressing these issues particularly important for ensuring accurate and relevant recommendations. In such communities, the influx of new users and items can significantly impact the system's ability to provide relevant recommendations, making it crucial to have an effective solution to both problems. By addressing the cold-start problem, this platform ensures that even in small and unique communities, users receive tailored and relevant recommendations.

These contributions collectively offer a novel solution that enhances user engagement, personalization, and the overall quality of the travel planning experience.

The article is structured as follows: it explores the context of software tools to enhance sustainable tourism in disadvantaged areas, it illustrates the most common AI techniques used in developing travel recommender systems, it describes the design and architecture of the B4R Platform, including its data sources, algorithms, and user interface. The article discusses the platform's features and functionalities, highlighting its innovative approach to personalized recommendations and storytelling. Results of a pilot study conducted to evaluate the platform's effectiveness in promoting tourism in the Marche Region are then discussed. The article concludes with

a discussion of the challenges and opportunities of implementing such a platform in the context of marginal territories, covering issues related to data availability, user engagement, and sustainability, and suggesting directions for future research.

2 Literature review

Travel is not only a means of personal exploration but also a vital contributor to the economic prosperity of communities and the generation of employment opportunities for both local and global businesses. As travelers embark on journeys departing from their origin cities, they visit and bring wealth to various Points of Interest (POIs) they wish to experience along the way (Li et al. 2016). The need to meet all of users' requirements, such as travel times, accommodations, and expenses, has ushered in a growing demand for "smart travel". Smart travel allows tourists to navigate a vast array of available POIs, carefully selecting those that align with their personal interests or those that promise new and exciting experiences, such as participating in local events and savoring local food, both of which significantly impact the overall tourist experience (Lu et al. 2021).

In response to the need for simplifying the process of crafting the ideal travel itinerary, a multitude of Recommender Systems (RSs) have emerged (Ricci 2022). These RSs have been specifically designed to assist tourists in the selection of the most suitable travel plans, considering their expressed preferences and their individual tourist profiles (Borràs et al. 2014) (Chaudhari and Thakkar 2020). Furthermore, contemporary studies, such as the work conducted by (Buhalis and Sinarta 2019) and (Esper et al. 2025), highlight the simplification of the planning process, made possible by ubiquitous internet access through mobile devices and cutting-edge connectivity technologies, including cloud services, thereby granting travelers access to information and services from virtually any location.

2.1 Travel recommender systems functionalities

The field of travel recommender systems has witnessed rapid and continuous evolution, driven by advances in machine learning, data integration, and user experience design. Traditional systems have long focused on core functionalities such as itinerary generation, destination suggestion, comprehensive tourist package compilation, and attraction ranking (Ravi and Vairavasundaram 2016).

Recent studies show that travel recommender systems have evolved dramatically over the past few years by integrating advanced machine learning techniques, sophisticated data integration, and enhanced user experience design to deliver highly personalized, context-aware suggestions. For instance, a comprehensive review by (Song Ke Xin et al., 2024) highlights that modern tourism recommender systems now leverage real-time geospatial data and embedded mapping services, such as Google Maps, to generate dynamic itinerary recommendations that are both personalized and location-specific. In addition, cutting-edge approaches like BTRec (Ho et al. 2023) use state-of-the-art language models to capture user behavior and trajectory information, enabling the generation of alternative itineraries that adapt to users' individual

preferences. These recent advances mark a significant shift from earlier systems that primarily focused on static itinerary generation and simple destination suggestions, underscoring the role of deep learning and contextual data in revolutionizing the travel recommendation domain.

DeepAltTrip, presented by (Rashid et al. 2023), focuses on generating top-k alternative itineraries. Their framework not only emphasizes popular routes based on historical travel data but also encourages diversity, ensuring that users are offered multiple viable travel routes. A self-supervised representation learning framework for trip recommendation has been proposed by (Gao et al. 2022). Their work highlights how self-supervised techniques can learn robust representations from sparse travel data, which is crucial for developing accurate route recommendations in environments where explicit feedback is limited.

Other recent developments have seen the emergence of holistic tourist pack systems that integrate multiple facets of the travel experience, including accommodation, dining, cultural events, and local transportation, using sophisticated scoring and ranking algorithms that optimize recommendations based on user preferences, diversity, and cost efficiency (Benouaret and Lenne 2016). The incorporation of social features has further enriched these systems. Platforms now routinely integrate user-generated content, ratings, and interactive notifications, which not only improve recommendation accuracy but also enhance user trust and engagement through community-driven insights (Jiang et al. 2016; Liu 2023).

However, contemporary challenges extend well beyond the refinement of recommender algorithms. Fragile and marginalized territories face socio-economic decline and cultural erosion, calling for innovative approaches that transform tourism into a tool for regional revitalization. Recent literature emphasizes the need for recommender systems to address ethical, sustainable, and community-centric dimensions (Banerjee et al. 2023; Bergemann and Bonatti 2024). In this context, modern platforms are challenged to integrate local narratives, cultural assets, and stakeholder profiling into the recommendation process. Such integration not only facilitates personalized travel planning but also supports the strategic reactivation of territories by promoting regional branding and local identity.

Our approach exemplifies these recent advances by embedding functionalities that extend beyond the traditional scope of tourism apps. The platform integrates interactive social features, such as dedicated “Stories” sections and networking tools, which enable local stakeholders to co-create and share authentic experiences. This collective narrative helps reinforce the unique cultural and natural identity of fragile territories, thus contributing to regional resilience. Moreover, by incorporating advanced profiling mechanisms that match user preferences with local service offerings, the system addresses key challenges such as the cold start problem and the filter bubble effect (Wang et al. 2023; Roy and Dutta 2022). Additionally, the growing body of research on sustainable tourism has underscored the importance of aligning technological innovations with broader socio-economic objectives. Recent studies (Nwaimo et al. 2024; Liu 2023) have demonstrated that recommender systems can act as catalysts for socio-economic development when they facilitate the convergence of personalized travel experiences with region-specific branding. In this light, our platform not only offers tailored itineraries but also serves as a dynamic tool

for the regeneration of marginalized areas, an imperative that remains largely unaddressed by conventional tourism apps.

2.2 Recommender system paradigms

The key expectation from a recommender system is to minimize the information overload by providing only relevant suggestions to the end user. Content-based (CB) systems, such as the one presented by (Van Meteren and Van Someren 2000), recommend items similar to the ones that users already liked in the past. The recommendation process is mainly focused on defining an appropriate measure to compare a user and an item (Borràs et al. 2014). When the items and the users are described by a list of keywords, some similarity measures can be applied. For example, in (Zhang et al. 2021) items and users are described using concepts from an ontology, which defines archetypes of tourists (e.g., cultural, sportive or adventurous), and the cosine similarity between the two vectors (user and item) is calculated.

However, recommending only items that are like those that a user has already interacted with, can lead to a filter bubble effect, where users are exposed to a narrow range of content aligned with their existing preferences (Wang et al. 2023). In this context, it is unlikely that the user will be offered items that do not fall within this range of preferences. To address this, the algorithm may incorporate techniques such as ontological similarity to recommend diverse and unexpected items to users (Kuznetsov and Kordík 2023), and category-based diversification.

Demographic-based (DB) filtering systems (Safoury and Salah 2013) rely on the user's demographic data, such as geographic region, age group, gender, season, etc., as common demographic classifiers. Users are assigned to a certain stereotypical class depending on their demographic data. Each stereotype provides an accurate description of what tourists want and how they behave in different situations. Although the DB approach does not require historical data on user transactions, it requires market analysis to identify demographic correlations between user groups and specific stereotypes. This analysis is normally performed by analyzing user profiles either from the organization database or by carrying out an explicit survey among users.

Collaborative filtering (CF) (Schafer et al. 2007) idea is that multiple individuals sharing similar interests tend to have inclination towards similar items from other areas as well. The similarity between users is figured out based on their past behavior. Common similarity measures being used by collaborative recommender systems include cosine coefficient and Euclidean distance (Renjith et al. 2020). It's essential to address certain weaknesses in CF systems (Roy and Dutta 2022). Data sparsity occurs when the number of ratings from users is small in comparison with the total number of items, so that the probability of finding users that rate the same items is too low to make good estimations. Finally, the CF approach also suffers from the scalability problem if the community of users is large (Borràs et al. 2014).

A pervasive challenge affecting nearly every recommender system is the cold start problem, which occurs when there is insufficient historical data to generate accurate recommendations. This issue manifests not only when a new user enters the system with little to no prior interaction but also when a new item is introduced without any ratings. To address this challenge, various solutions have been proposed in the litera-

ture. For instance, Graph Convolutional Networks (GCNs) propagate information through the user–item graph, enabling the system to infer preferences for both new users and items by leveraging the relationships across the entire network. Transfer Learning techniques help by identifying overlaps in user preferences across different contexts, while Active Learning approaches involve soliciting user feedback to quickly gather necessary data (Roy and Dutta 2022). Moreover, hybrid recommender systems combine the strengths of multiple approaches, and various strategies for aggregating these methods have been developed to further mitigate the cold start challenge (Isinkaye et al. 2015).

The literature on recommender systems highlights a significant research gap concerning the integration of cold start solutions with context-aware strategies, particularly for tourism in fragile territories (Bawack et al. 2022).

2.3 AI techniques in travel recommender systems

Over the past decade, a wide range of advanced AI techniques have been employed to enhance recommender systems, resulting in recommendations that are both more accurate and more finely tuned to users' interests (Nwaimo et al. 2024). Recent approaches leverage optimization methods, clustering algorithms, and sophisticated knowledge representation techniques to capture complex user preferences and item attributes. For instance, modern clustering methods group users based on demographics, behavioral patterns, or explicit ratings, thereby enabling the system to recommend services that have been positively evaluated by similar users (Zhang et al. 2024). When users and items are each represented as vectors, vector-based classification methods, such as those calculating cosine similarity or Euclidean distance, are used to assess their proximity, which helps determine the most suitable services for specific user groups (Lee et al., 2023). Additionally, traditional algorithms continue to be refined: Support Vector Machines have been enhanced through advanced kernel methods to better classify user preferences (Kim et al. 2023), while the k-Nearest Neighbors algorithm remains effective for identifying similar users, as recently demonstrated in a movie recommender system that integrated k-NN with K-Means clustering (Kim et al. 2023). Moreover, approximate reasoning methodologies, including fuzzy logic (Gupta et al. 2018) and Bayesian networks (Hsu et al. 2012), have been adapted to manage the inherent uncertainty in user preference inference and to estimate the likelihood that a user belonging to a particular group will favor certain items (Nwaimo et al. 2024; Chen et al. 2023). By seamlessly integrating these diverse AI techniques, contemporary recommender systems are now capable of delivering tailored solutions for various applications, ranging from route recommendation to social recommendation platforms, thereby enhancing overall user engagement and satisfaction.

2.4 Research gaps and contributions

Despite the growing body of work on travel recommender systems, no existing platform applies recommendation techniques to deliberate place-branding for touristic

cally fragile areas, such as the small villages of the Macerata hinterland analysed in this study.

Although many hybrid approaches combine content-based and collaborative filtering to improve recommendation accuracy, most of these models focus on traditional e-commerce environments and do not adequately address the unique challenges associated with sparse data and limited user interactions in underrepresented regions (Chouiref and Hayi 2022; Luo et al. 2021). For instance, while recent studies (Nwaimo et al. 2024; Zhang et al. 2024) have proposed adaptive mechanisms to tackle the cold start problem, these approaches often overlook the critical role of local cultural narratives, socio-economic dynamics, and stakeholder involvement.

This gap indicates a need for a recommender system that not only integrates diverse data sources to mitigate cold start issues but also leverages region-specific contextual information to enhance personalized travel planning and promote sustainable regional revitalization. Our work aims to fill this gap by developing a hybrid recommender framework that explicitly incorporates local contextual factors into its algorithmic design, thereby providing more accurate recommendations for fragile territories.

3 Design and development of the proposed platform

The primary goal of the B4R Platform is to gather insights and data from analyzed regions, facilitating collaboration and connections between supply and demand while creating personalized experiences for interested users. Service providers, such as tourist operators, institutions, and local communities, will offer these experiences. The platform cultivates stronger branding, focusing on identifying or establishing a shared identity rooted in the hidden qualities of a place, resulting in greater visibility (Schröder 2018). The B4R platform aims to revolutionize territorial branding by examining scenarios and patterns to structure the region's offerings (e.g., mobility, cultural sites, events, and tourist attractions). It also seeks to develop suitable storytelling techniques for communicating different visions in a personalized manner to various user profiles, defining customizable visit routes. The system tailors these routes to each visitor's profile (age, gender, geographical origin, etc.), preferences, and needs (e.g., length of stay, interests, reasons for visiting, family status) using artificial intelligence algorithms that map the city/region's offerings to the user's profile (cluster of belonging). Determining the services to offer and the platform to use was essential. We designed and implemented a collaborative web platform aimed at enhancing the tourism experience by co-creating highly personalized itineraries.

3.1 Platform services

During the Discovery Phase, the platform's core functionalities were deliberately shaped by pressing challenges identified in contemporary tourism research. The socio-economic decline observed in many fragile territories called for features that could directly bolster local visibility and economic activity. Consequently, core functionalities such as a comprehensive search engine for underrepresented ser-

vices, a stakeholder networking module, a personalized itinerary planner to encourage sustainable exploration, and a community “stories” section were prioritized to address these needs. We chose each feature to align with specific challenges with specific challenges: for example, the networking and storytelling components integrate local narratives and cultural assets into the tourist experience, actively involving local stakeholders in content co-creation and thereby strengthening community engagement.

These identified functionalities were further refined through a rigorous analysis of best practices and competitor platforms, ensuring that the design learned from state-of-the-art solutions. This process highlighted effective features used by leading tourism services – for instance, the use of profile-driven recommender engines and interactive mapping of attractions, as seen in platforms like TripAdvisor (Tripadvisor, 2025), and the inclusion of local storytelling sections to engage communities, as exemplified by the Venezia Autentica italian project (Venezia Autentica, 2025). At the same time, the competitor analysis revealed critical gaps in existing offerings: most current solutions failed to emphasize sustainable tourism or to implement the kind of user and stakeholder profiling needed to match local offerings with visitor interests. These insights directly influenced our design decisions, reinforcing the integration of personalization and community-centric features that align with sustainable tourism principles.

The platform’s functional design has been guided by a strategic goal: exploiting innovative technology to address these territorial challenges and foster sustainable regional development. Each functionality is aligned with this goal, from stimulating socio-economic revival in marginal areas to preserving local culture through community engagement. This approach reflects the project’s overarching vision of co-designing digital solutions with local communities to revitalize fragile regions. By doing so, the platform not only promotes the cultural and natural heritage of these areas but also contributes to reducing regional disparities through a more equitable and sustainable tourism model.

A key priority within the B4R project was mapping the territory’s offerings with user preferences and characteristics, which led to the integration of an embedded map to represent services offered by each community and stakeholder. We designed the search engine to show a list of diverse services from each entity (Benouaret and Lenne 2016), such as cities, hotels, and restoration points, ensuring a variety of recommendations. However, unlike the system developed by (Benouaret and Lenne 2016), we chose not to design ours as a tourist package recommender system for two main reasons. First, we decided to allow users to compile their own packages from a list of services, giving them the freedom to include multiple items from the same entity. Our goal is to ensure that the recommendation list prominently features the items the user is most likely to select or include in their package. Second, the platform is designed to cater to the networking needs of our stakeholders, not just as a recommender engine. This is why we are working towards integrating social aspects into the platform. We designed our platform to be a route recommender system, using technologies like Google Maps Services, and allowing users ultimate choice over what POIs to visit. Furthermore, according to (Kurata 2010), users complained about the inability to customize the recommended tour plans by adding or removing POIs

which they want to visit/avoid. Therefore, we did not implement optimization algorithms; instead, we allow the user to drag and drop the POIs in the Google Map frame and change the order of the sequence in which they intend to visit them. We do provide an initial suggestion by separating the services they can perform in the morning, afternoon, and evening, and by alternating them with restaurants and places to rest among the ones they chose from the list.

Another priority was to facilitate the use of our platform during the planning stages of a trip. Consequently, web-based interfaces emerged as the optimal choice for our recommender system, which incorporates an embedded map. This design provides users with easy access from any internet-connected device, eliminating the need for downloads or installations, and enables seamless exploration of attractions, driving routes, and approximate visiting times. Our platform supports a dual role for users, allowing them to function as both providers and consumers based on their skills and needs. Providers can create new services, such as organizing events, enhance their networking opportunities, and gain visibility. Consumers, on the other hand, can participate in events, create itineraries, and share stories, comments, or pictures. Additionally, social features are integral to our vision of a platform that not only connects users but also intelligently collects data from designated focus areas. These features aim to foster a dynamic ecosystem where users actively contribute to and enrich their travel experiences.

In summary, our platform offers personalized driving routes while integrating social features to create a dynamic environment. This approach supports both the connection of users and the intelligent collection of data, with the aim to enhance the overall travel experience.

3.2 User profile representation

To create a tour of attractions relevant to the user, our system relies on profiling the user's preferences. In the literature, the simplest model to represent the user profile associates to each user a list of preferred keywords or categories in which items are pre-classified. Another approach to represent the user profile involves associating each user with a numerical ratings vector corresponding to the attributes of the POIs, indicating the users' degree of interest (Garcia et al. 2011). Alternatively, semantic models, utilizing ontologies, provide a structured representation of user preferences (Wang, Zeng, Tang; 2011). For example, this keyword list may contain names of classes in the ontology (i.e. concepts) and the ratings vector can also be defined in the space of the concepts in the ontology, while (Kurata; 2010) in his research work has estimated the expected utility of POIs based on the vector matching of the POI's characteristics and the user's profile. In moreTourism (Rey-López et al. 2011) the users have an associated tag cloud with terms relevant to their profile, and a new tag is created for each attraction based on the tags of the users who liked it. The system uses this information to compare tag clouds of users and items and find coincidences.

In adherence to a design thinking approach, a collaborative "Discovery Phase" was conducted to identify the functional requisites of the B4R Platform and establish personas used to envision user scenarios. We developed user profiles that summarise primary characteristics, desires, and features the platform could offer in response

to their expressed needs. Specifically, the Discovery Phase involved constructing a detailed “Value Map” for the traveler persona, identifying critical aspects such as desired emotional and social outcomes, typical tasks, and encountered difficulties. From this analysis emerged two key dimensions for user modeling: “Themes” and “Moods.” “Themes” encapsulate the traveler’s interests, such as cultural, gastro-nomic, natural, and innovative experiences, while “Moods” reflect the social and emotional contexts of travel, including traveling alone, as a couple, in groups, or with family. We directly derived these concepts from the user pain points such as difficulties in itinerary planning, limited accessibility of localized information, and challenges in finding authentic, emotionally impactful experiences. Consequently, this targeted profiling facilitates the generation of personalized recommendations, effectively matching user expectations and enabling meaningful interactions with local communities, thereby addressing specific functional and emotional requirements of travelers identified during the Discovery process.

3.3 Preferences over ratings

Contrary to the approach outlined by (Wang 2023), who analyze the factors affecting users’ interest preferences in terms of their global and local rating information, the B4R project deliberately avoids the use of rating information, steering clear of a system reminiscent of platforms like TripAdvisor. Other papers in literature employ alternative approaches rather than relying on ratings. For instance, (Asaithambi et al. 2023) propose a model that integrates analytics of a variety of big data, such as images, reviews, climate, social media, and location, while (Lorenzi et al. 2011) discuss an “assumption-based multiagent” system that makes travel package recommendations using user preferences.

The decision to forgo ratings is substantiated by various reasons highlighted in the literature, including concerns related to fairness and data limitations. Recommender systems that use ratings can sometimes lead to unfair outcomes, such as over-representing popular destinations or under-representing less well-known ones contributing to problems such as “*overtourism*” and “*undertourism*” respectively (Banerjee et al. 2023). Another equally important motivation was to reward those who aren’t skilled at self-promotion, the B4R project is particularly attentive to this concern. Moreover, ratings-based recommender systems require a large amount of rating data to function effectively. However, in many cases such as ours, such data may not be available or may be sparse, leading to less accurate recommendations (Roy and Dutta 2022) (Wang 2023). Therefore, we chose not to rely on user ratings, but on the latent factors that might drive the ratings, ensuring a more dynamic and inclusive approach to recommendations.

In response to this, our platform strategically categorizes services into two functional groups: “attractions” and “services and facilities” (Gunn et al., 1980). This distinction aims to achieve a more precise matching between users’ preferences and the available offerings. By organizing the platform’s content in this manner, we enhance the user experience by presenting recommendations that are closely aligned with specific preferences. Additionally, this categorization allows for the creation of tourist package-like recommendations by alternating activities with places to rest during the

initial itinerary creation. Users can then modify these suggestions to better fit their individual needs.

3.4 Platform development

The analysis of the platform services led to a list of the main functional modules that can be represented and detailed through the architecture of the platform.

3.4.1 Territory services & info and custom itinerary

After logging in, users are directed to the platform's search engine showed in Fig. 1, which enables them to browse available services and products that can be incorporated into personalized itineraries. The custom itinerary feature is an essential aspect of the platform. Using Google Maps integration, users can visualize and modify their routes according to their preferences. The system suggests a base itinerary based on user profiling and the services offered by providers in a specific region. Users can then customize these routes further by adding or removing destinations, ensuring that each itinerary is tailored to their unique interests.

3.4.2 Territory services and info insertion form

Providers have access to a dedicated section where they can input all relevant information about the services and products they offer. This process is facilitated through a structured insertion form, which collects data on various offerings such as local products, events, accommodations, and more. This information is then made available to consumers for use in their itinerary planning. The form ensures that all necessary details are included, making the interaction between providers and consumers efficient and seamless.

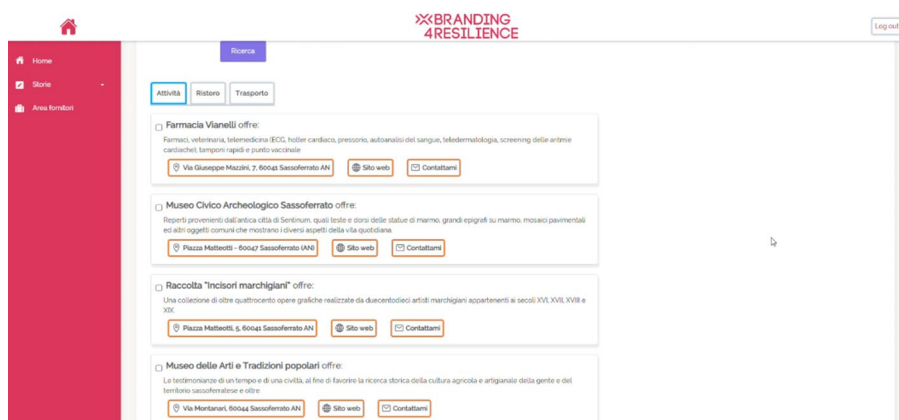


Fig. 1 The services and info search engine

3.4.3 Providers section

A key feature for providers is the ability to network with other users in the same category. This is achieved through a messaging system that supports both asynchronous and synchronous communication. Providers can collaborate, share information, and build a network of partnerships that benefit both their businesses and the wider community. This feature enables them to coordinate initiatives, exchange knowledge, and address common needs, thereby fostering a collaborative ecosystem within the platform. Lastly, providers are given the ability to create and personalize pre-developed tourist itineraries. These pre-set routes are available to consumers, who can further customize them based on their preferences. Providers can adjust these itineraries to highlight their services, ensuring that their offerings are well-represented and accessible to potential visitors. This dynamic interaction between consumers and providers through customizable itineraries enhances the overall user experience and promotes a more tailored approach to tourism in the regions targeted by the Branding4Resilience project.

3.4.4 Recommender system handler

The Recommender System Handler is responsible for managing the Machine Learning model that generates statistical matches between supply and demand. This module operates independently from the rest of the platform, is fully implemented in Python, and functions as a service that runs periodically to retrain the model based on the data stored in the database. This ensures that the model remains up-to-date with new inputs provided by both Providers and Consumers through the Experience Platform. The Recommender System Handler interacts with the platform by accessing the database, where it writes the results of each new prediction, and may also utilize APIs if required for client-side requests. Additionally, a web crawler can be employed when necessary to gather external information from the web, enhancing the dataset and improving the model's prediction accuracy.

3.5 The recommender system

3.5.1 Data collection

The user profile representation is constructed through a dual approach of explicit and implicit data collection. Explicit preferences are gathered during the registration process. To maintain a user-friendly experience, we have streamlined the data input process by minimizing intrusive questions. At registration, users are presented with an animated form, shown in Fig. 2, that clarifies the data collection process and how their information will be handled in compliance with the European GDPR normative. Users can input their demographic data (age, gender, occupation) and preference data such as preferred theme of affiliation, preferred means of transportation to use during the route suggested by the recommender system, likes and dislikes in the form of tags, and other preferences in a free text field. Data is both numeric and categorical in nature, due to the inclusion of “themes”, “moods”, and “likes/dislikes” that

Fig. 2 Registration form where users can input their preference data such as preferred theme and means of transportation, tags, and other preferences in a free text field

represent the preferences of the users. Analogously to (Kurata and Hara, 2013), our platform supports two types of user profile (theme and mood), and the registration form includes radio buttons where users can select their preferred attraction type.

“Theme” refers to the tour’s functional feature demanded by the user. We considered six categories based on the Discovery Phase that led us to understand the Sas-soferrato area in terms of available offerings: culture and art, handicraft and tradition, sports and entertainment, food and beverages, nature and adventure, and innovation.

- Culture.
- Handicraft & Tradition.
- Sports & Leisure.
- Food & Beverages.
- Nature.
- Innovation.

“Mood” refers to the tour’s emotional characteristics demanded by the user. We considered five types of mood, each with an appropriate tag representing the numerosity of the group: couples for romantic, family for kids, group for fun, alone for peace, and work for productivity. We also expect mood to be relatively stable and consistent, while theme may change depending on the destination and context.

By integrating these categories into our travel recommender system, users can receive personalized recommendations tailored to their general area of interest.

Simultaneously, implicit preferences are collected as users interact with the platform, capturing insights into their behaviors and preferences that we use to enhance the recommender engine. Additionally, we collected data regarding the offers in the Sassoferrato area to initialize the services available on our platform. This data, sourced from Excel tables kindly provided by a local association, incorporates information collected from institutions, services, and various stakeholders regarding tourist companies, craftsmen, educational and cultural institutions, agri-food companies, innovative technological companies, associations and local action groups (GALs). Even more data can be created on the platform by providers. The platform provides these users with the “Territory services and info insertion form” module where they are able to create new venues, events, and or services that other users of the platform can partake into. Only platform administrators can approve new services created on the platform after a careful manual review. The creation forms have “theme” and “tags” fields for the fast categorization of the created venue/event/service. This allows the recommender engine to be able to match the newly created offer with respect to the users’ themes, likes and dislikes, even in the absence of direct interactions and immediately suggest it if it were relevant.

3.5.2 Recommender engine

The recommender engine, implemented by the Recommender System Handler module, is a crucial component of the B4R web platform. The engine utilizes a combination of fuzzy searches, content-based, and collaborative filtering approaches to provide personalized recommendations. At its core, the engine calculates cosine similarity between vectors representing users and services, helping users sort relevant offerings from various entities, such as products, hotels, or restaurants. This process not only suggests suitable attractions but also assists users in crafting tailored itineraries by mixing and matching multiple possibilities. To further enhance search capabilities, we incorporated an Elasticsearch-based engine, allowing keyword-based searches and enabling fuzzy matching to broaden the scope of relevant options. In this way users can easily find the offerings and experiences that align with their interests. This integration increases diversity of the results by including services with similar characteristics before computing their similarity with the user’s preferences.

The Recommender System Handler is composed of several sub-modules, each responsible for specific functions of the engine, i.e. compute the cosine similarity, sort services accordingly, and refresh the recommendation list with newly created or previously unnoticed services, ensuring that users always have fresh and relevant choices. The cosine similarity measure has been chosen for its emphasis on frequency, aligning well with our data. This measure is most appropriate for data where frequency is more important than absolute values, notably, term frequency in documents (Connor 2016) (Heidarian and Dinneen 2016). This makes it particularly suitable for our use case, since we are primarily interested in the frequency of interactions between users and items. One of these sub-modules handles communication with an Elasticsearch server, optimized for search engine-based tasks, to process users’ search queries and quickly output items with similar characteristics. Another crucial sub-module is responsible for computing the similarity between users and the

services obtained in the previous step. The similarity is calculated using a similarity matrix, which relies on vectors representing users and services. Analogously to the approaches taken by (Garcia et al. 2011) and (Kurata 2010), we represent the user profile by associating each user with a numerical ratings vector. Each user and service are encoded as a vector, that encapsulates both inherent attributes (such as demographics for users or categories for services) and interaction-derived data, in a multi-dimensional space, where each dimension corresponds to a specific feature, i.e. the attributes of the services associated with the users' degree of interest. Initially, both user and service vectors are constructed using direct information (e.g., user demographics, service attributes). In this scenario, a user vector u_i and a service vector s_j , are defined as:

$$u_i = [u_{i1}, u_{i2}, \dots, u_{iN}] \quad (1)$$

$$s_j = [s_{j1}, s_{j2}, \dots, s_{jN}] \quad (2)$$

where u_{ik} and s_{jk} represent the values of the k -th feature for user i and service j , respectively. As users interact with the platform by selecting services, these interactions are recorded and used to update the vectors, ensuring that recommendations remain accurate and personalized over time. Each time a user selects a service, the corresponding feature frequencies in both the user and service vectors are updated. In this way, when a user interacts with a service (e.g., selects a cultural event or books a nature-related activity), the system records the interaction and adjusts the corresponding attributes in both the user's and the service's vectors. For example, if users consistently choose services tagged as 'Cultural,' the weight of this category in their vector will increase, and the system will start recommending more services of this type to similar users. This dynamic update process not only personalizes recommendations for individual users but also refines the service's profile, making it more relevant for other users with similar preferences.

This process can be represented mathematically as follows:

$$u_i \leftarrow \eta \times s_j + (1 - \eta) \times u_i \quad (3)$$

$$s_j \leftarrow \eta \times u_i + (1 - \eta) \times s_j \quad (4)$$

Where η is the learning rate, $1 - \eta$ is the decay rate, controlling how much the interaction between user u_i and the service s_j affects the vectors with respect to the previous values of the users or services, respectively. This is a convex combination that preserves the range of the vector components. The updated vectors u_i and s_j now include not only the original static data but also a dynamic component reflecting the frequency and nature of interactions. The interaction frequency is crucial in refining the vectors over time.

Following (Isinkaye et al. 2015), Cosine Similarity has been used as a metric to measure how similar user and service vectors are. This similarity score ranges from

– 1 to 1, with higher values indicating a closer match between the user’s preferences and the service’s attributes. However, since the values are derived from frequencies, all vectors lie in the positive quadrant of the N-dimensional space, with cosine similarity values ranging from 0 to 1.

Categorical data, such as service features, are encoded using one-hot encoding, which allows effective similarity computation even with limited interaction data.

In the absence of interactions, a sub-module employs the k-Nearest Neighbors (k-NN) algorithm to initialize the features of users and services by averaging values from the most similar users or services, respectively. For example, a new user may be recommended services chosen by users with similar demographic profiles, while a new service might be suggested to users who have selected items with similar characteristics. This is a hybrid approach, combining content-based (CB) and collaborative filtering (CL) methods, however, instead of switching between these methods based on specific conditions, the engine weights the results, giving more significance to direct interactions when computing similarity. Collecting direct and indirect information from the user and recommending items accordingly helps address the cold-start problem (Roy and Dutta 2022). The weighting was done also to address the limitations of the CB part of the algorithm which tend to recommend items that are similar to those that a user has already interacted with or expressed a preference for which can lead to a filter bubble effect (Han et al. 2022).

The engine then sorts and presents the results, categorizing them into “attractions” and “services and facilities”, with recommendations ordered by similarity coefficients. The platform offers more recommendations than necessary, allowing users to select and customize their itineraries according to their preferences, without strictly adhering to what the engine suggests. Users can view and modify their selected Points of Interest (POIs) on the embedded Google Map, adjusting routes, adding or removing POIs, and selecting transportation modes such as biking, driving, or walking. When a user interacts with the Google Map, such as viewing selected items or checking the itinerary’s time and length, the platform registers this interaction. Each interaction is recorded and used to update the features for future recommendations. Additionally, the engine proposes an itinerary that balances activities throughout the day with appropriate rest periods. The user retains full control over final decisions, customizing the plan by adding or removing POIs.

The final step in the recommendation process involves updating the features associated with users and services. Each interaction, along with all the recommended and chosen services, is then sent to the Elasticsearch database. When a user interacts with a service, the characteristics of that service are updated. This update incorporates data from the new user interaction, reinforcing the relationship between the users and the service. Over time, this process allows each service to accumulate data not only about its inherent characteristics but also about the users most likely to interact with it, facilitating cross-recommendations. This update process is akin to how Instagram influencers influence their followers, where the first users who interact with a new service are the “influencers”. These influencers are the early adopters, the ones whose preferences and behaviors impact the platform’s understanding of the service. When these influencers interact with a service, their preferences, represented by their features, are updated first, reinforcing the relationship between them and the service. As

a result, the service begins to “learn” about the types of users most likely to engage with it. Over time, the interactions of these early adopters influence the features of the services they interact with. This, in turn, affects what other users, the “followers”, see in their recommendation lists. Followers might start seeing different services recommended to them or see the same services ranked differently based on the influencers’ interactions, shaping the content for the users of the platform with their interactions. This dynamic helps the platform adapt and refine its recommendations, ensuring that they are more aligned with user interests and preferences as the system evolves.

One of the key strengths of this approach lies in its continuous learning process. Each interaction updates the vectors, ensuring that the similarity computations reflect the most current data. This ongoing adaptation enhances the accuracy of the recommendations, allowing the system to evolve alongside the users’ preferences. However, this continuous update process can lead to the convergence of feature values toward an average, a phenomenon that may exacerbate the filter bubble effect. As the feature values converge, the distinctiveness of individual user profiles and service attributes diminishes, potentially leading to a homogenized recommendation list where only similar items are suggested. This is particularly evident when services are recommended randomly, as random feature values mix more readily, accelerating the convergence process. This behavior is less pronounced when recommendations are based on higher cosine similarity scores, as the features are only updated by users who genuinely prefer those services. However, high similarity interactions yield milder updates.

To mitigate the filter bubble effect, we implement two key strategies: normalization and the periodic introduction of new or previously unused services into the recommendation lists, ensuring a diverse range of options. Normalization of the features during the update process helps prevent excessive convergence, which helps maintain diversity in the recommendations. Additionally, by periodically introducing new or unused services, the platform ensures that users are exposed to a broader range of options, thereby reducing the likelihood of a filter bubble forming. In line with research by (Savir et al. 2013), which advocates for balancing diversity in recommendations, our platform enhances diversity in two primary ways: by suggesting newly created services and by promoting services that have not been used in a set number of interactions. The system recommends new services to users whose preferences align with similar offerings, determined through a k-NN clustering approach. Unused services are randomly suggested to users to increase their visibility, even if they have not met a minimum rating threshold. This approach potentially attracts users with different interests, broadening the range of interactions. Rather than imposing a strict similarity threshold, the platform uses a probabilistic approach, setting a parameter for the percentage of recommendations dedicated to new or unused services. We take a probabilistic approach to introduce new elements in the recommendation to avoid the risk of replacing the entire top-k items list with only new or unused items, depending on their numerosity. The assumption is that this controlled randomness introduces greater variability in feature updates and broader visibility in subsequent interactions. Users are encouraged to apply filters in the search form to prevent services unrelated to the type of itinerary they want to build from appearing in the recommendation list.

In summary, our platform's recommender engine takes a comprehensive approach, blending user interactions, demographic data, and continuous feedback to deliver a more personalized travel planning experience. This integration of multiple data sources and dynamic interaction analysis contributes to the overall quality and flexibility of the recommendations. However, while the primary goal of this recommender system is to minimize information overload by providing relevant suggestions, we display all possible suggestions to users. This approach allows us to evaluate how users' choices rank among the full list and measure the sensitivity of our algorithm.

4 Experimental case study

The evaluation process was designed to assess both the subjective aspects, such as user experience, perceived usefulness and perceived innovation, and the objective performance metrics of the recommender engine.

4.1 User experience, perceived usefulness and perceived innovation evaluation

Users received an invitation to participate in a questionnaire, which was administered via Google Forms. During this step, the users had to accept a privacy policy to be compliant with the GDPR European privacy laws and complete a form with additional personal information. We also informed users of the purposes of the questionnaire itself, i.e. to assess the impact the platform could have on local tourism and to evaluate whether using the platform has sparked curiosity about the Marche hinterland. To achieve our objectives, we collected frequently asked questions from the literature (i.e. (Parasuraman et al. 1988) and (Davis 1989), both still widely used in the literature in various fields, such as in (Liestyanti and Prawiraatmadja 2021) and (Bano and Siddiqui 2024) and designed additional items specific to our platform. Data was collected in an aggregate manner to facilitate statistical analyses. The questionnaire started by asking how often they traveled for tourism purposes, and it continued with 18 5-point Likert scale questions to ascertain the following properties about the platform or the users using it:

1. *Frequency of use during travel (of our or similar platforms).*
2. *Frequency of use before travelling (of our or similar platforms).*
3. *Impact on discoveries (new knowledge on the focus territories acquired through the platform).*
4. *Desire to make a trip (how much the user would like to visit the places he discovered through the platform).*
5. *Improvement of experience (how much user's experience improved thanks to the platform).*
6. *Absence of issues (whether there are any issues with the platform use).*
7. *Planning utility (how much the user found the platform useful in the planning process).*
8. *Content accuracy (whether the content shown in the platform is up to date).*
9. *Industry-standard features (whether the user finds components and tools that are widely recognized and commonly integrated within modern platforms).*

10. *Lack of complexity (how much the user found the platform complex to navigate and use).*

11. *Intuitiveness (how much the user found the components and tools available easy to grasp the first time).*

12. *Content reliability (whether the content shown in the platform reflects real data).*

13. *Content presence (whether the content shown in the platform was sufficient to showcase the services).*

14. *Recommendable (how much the user thinks he would recommend the platform to his peers).*

15. *Travel preparation support (how much support the platform offered when the user is preparing to make a trip).*

16. *Cutting-edge (how much the user thinks there are innovative components and tools that push the boundaries of current technology).*

17. *Inspiring (how much the user would likely find inspiring the point of interest he wants to visit based on its description on the platform).*

18. *Endorsement (how willing users would be to support the development of the platform).*

The questionnaire received 67 responses. We conducted Cronbach's alpha to assess the internal consistency of the questionnaire.

4.2 Recommender engine evaluation

To evaluate the recommender engine, we employed the sensitivity metric. This kind of analysis is crucial in evaluating the effectiveness of the recommender engine in providing relevant suggestions. To compute sensitivity, we analyzed the interactions of users with the services during the platform's deployment. Sensitivity is defined as $\frac{TP}{TP+FN}$, the ratio of true positives TP to the sum of true positives TP and false negatives FN . If we assume that the user interacts and selects those services he actually considers relevant, then among all the services the user considers relevant and interacts with, some of them rank high based on the similarity score and are shown to the user by the recommender system (TP) and other services are not predicted as relevant by the recommended system and are not shown to the user (FN). Therefore, if the recommender system showed only the first top- k services after sorting by similarity, meaning that the engine classified the services which are not shown as negatives (either as true negatives TN or false negatives FN), we would be able to compute the sensitivity as a function of the number k of services shown to the users. A further experimental evaluation was conducted to compare the performance of our hybrid recommender system with a traditional rating-based collaborative filtering approach. The baseline recommendations were computed using the Singular Value Decomposition (SVD) algorithm from the Surprise library (Hug 2020), trained exclusively on user ratings collected during users' use of the platform.

This choice has been made because matrix-factorisation models still today remain the de-facto benchmark for rating-prediction tasks (Samra et al. 2024), while the Surprise library was selected due to its robustness, extensive use in academic

research, and its reliance on explicit user-item ratings to predict the relevance of recommendations.

To prepare data for this baseline, implicit interactions were mapped onto a 1–5 preference scale (1 = least preferred, 5 = most preferred). Missing ratings were implicitly estimated by the model during factorisation.

For a comparison with our hybrid approach, we evaluated both engines on Precision, Recall, and F1-Score at $k=20$, i.e. each user received a top-20 list from the SVD baseline and a top-20 list from the hybrid engine. Precision was defined as the fraction of recommended items within those top 20 that the user actually selected, and Recall was the fraction of all items the user selected that appeared in the top 20. The F1-Score was computed as the harmonic mean of Precision and Recall.

To capture recommendation diversity, we measured item coverage as the percentage of unique services recommended across all users relative to the total number of available services on the platform, i.e. counting how many distinct items from the full catalog appeared in any user's top-20 lists, and dividing this by the total catalog size. Novelty was assessed by identifying, in each top-20 list, those items that had not been previously recommended to any user during the training period, while the novelty score is the proportion of such “new” or rarely recommended items appearing in the recommendations. By computing item coverage and novelty for both the SVD baseline and the hybrid approach, we could quantify not only accuracy but also how effectively each method surfaced under-promoted services.

5 Results

This chapter presents the results of the various analysis conducted.

5.1 Questionnaire results for user experience, perceived usefulness and perceived innovation

Cronbach's alpha was the first and was used to assess the internal consistency of the questionnaire. The constructs included in the questionnaire were tourist platforms usage, utility, content quality, design and usability, and innovative elements. Cronbach's alpha with 95% confidence reveals that the internal consistency of the constructs is more than acceptable on average.

Table 1 shows a summary of the results obtained, and we are interested the most in the answers to the questions “Do you think that trying our B4R platform might have sparked your interest in visiting the Marche hinterland?” and “Does the B4R platform enhance your tourist experience in the area?” represented by the variables “*Desire to make a trip*” and “*Improvement of experience*”, respectively. While no maximum marks have been given, as shown in Table 2, indicating room for improvement, we are satisfied with the median results and believe we have successfully achieved our objective.

Table 3 shows a summary of the results obtained from the questionnaires, aggregated by the constructs. While minimum marks have been given, the median values show positive results and an overall satisfaction among all categories that we studied.

Table 1 Cronbach's alpha analysis

	Cronbach's alpha	CI 95%
Use of tourist platforms	0.63	[0.247, 0.855]
Usefulness of the platform	0.71	[0.158, 0.897]
Quality of the platform's content	0.51	[-0.053, 0.808]
User experience	0.55	[-0.275, 0.844]
Innovative elements of the platform	0.75	[0.271, 0.911]

Table 2 Question 4 and question 5 statistical indicators

	Mean	Std	Min	Median	Max
Q4	3.13	0.81	1	3	4
Q5	3.69	0.70	2	4	4

Table 3 The five constructs statistical indicators

	Mean	Std	Min	Median	Max
Use of tourist platforms	3.55	0.88	1	4	5
Usefulness of the platform	3.50	0.87	1	4	5
Quality of the platform's content	3.36	0.80	1	3	5
User experience	3.66	0.73	2	4	5
Innovative elements of the platform	2.97	0.92	1	3	4

Not all users were in agreement, which is reflected in the good to average values of the Cronbach's alpha analysis.

5.2 Recommender system evaluation results

Sensitivity analysis was conducted to evaluate the effectiveness of the recommender engine in providing relevant suggestions, as it helps determine how well the system captures and ranks the items that users are most likely to select. By varying the number of top-k recommendations shown to users and observing how often the items they selected appeared within those top-k lists, we were able to compute the sensitivity as a function of the number k of services shown to the users. The sensitivity of the recommender engine was calculated as the proportion of the user-selected services that appeared within the top-k recommendations. The resulting plot in Fig. 3 shows the average number of recommendations to show to users and a 95% confidence interval (CI) to achieve a certain sensitivity threshold.

To achieve 100% sensitivity for all users, i.e. where every service selected by each user is included in the top-k recommendations, an average of 27.25 recommendations is required. The 95% CI for this average range from 21.37 to 33.14 recommendations. This means that if the recommender engine suggests a list of 28 items, all of a user's preferred services would appear in that list 95% of the time. In contrast, if recommendations were presented in a random order rather than ranked by the cosine

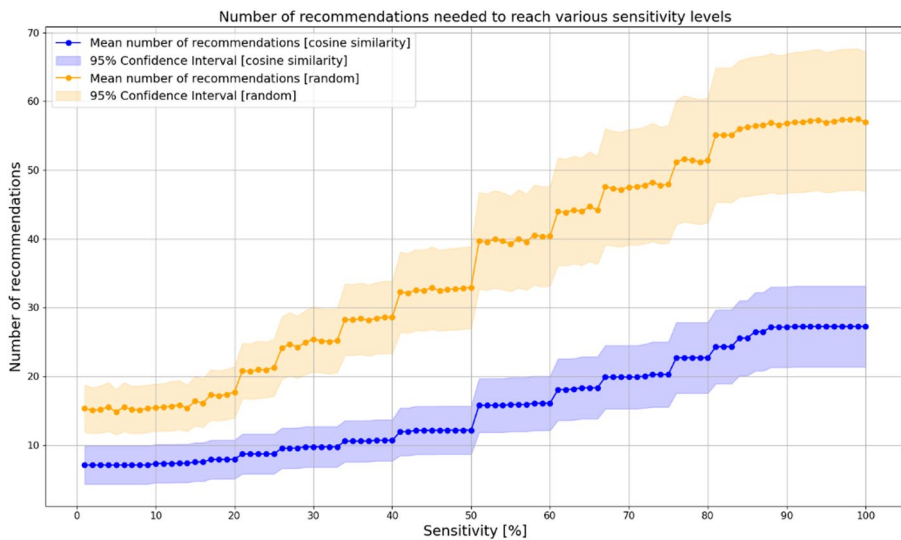


Fig. 3 Number of recommendations to be shown to users and 95% confidence intervals to achieve a fixed sensitivity. Ranking suggestions by user-service similarity (blue) outperforms a random suggestion model (yellow), indicating that shorter recommendation lists are sufficient

similarity measure employed by our system, users would need to sift through an average of 57.00 recommendations to find all their preferred services. The 95% CI for the random scenario ranges from 46.81 to 67.19 recommendations, highlighting the inefficiency of random ordering. This significant difference underscores the superior performance of our recommender engine compared to a random baseline, effectively validating its ability to provide relevant and accurate suggestions.

Three important factors should be noted. First, user interaction patterns significantly impact the number of recommendations required to achieve full sensitivity. For example, if a user frequently selects 10 or more items to include in their itinerary, the total number of recommendations needed to capture all their preferences within the top-k list will naturally increase. For instance, if a user interacted with 20 items, it would be physically impossible to have all 20 items selected by a user appear within a top-3 or top-19 list, thereby necessitating a higher k value in such cases. Second, our ability to compute sensitivity relies on having access to the complete set of user preferences. This assumption, a crucial concept in economics, presumes that the set of preferences includes all and only the services a user interacts with when creating an itinerary. While we acknowledge that a user's preferences may vary between interactions, we assume that the services he/she interacts with at each point in time form the complete set of his/her preferences for that specific interaction. Third, we computed the confidence intervals presented here with the t-distribution, assuming that the number of recommendations required to achieve a fixed sensitivity threshold follows a normal distribution. However, the data likely follows a Weibull distribution. Indeed, the number of recommendations cannot be negative, and the last recommendation may be infinitely postponed by the user (like endlessly scrolling through the YouTube app when searching for a video to watch). While one might assume that the number

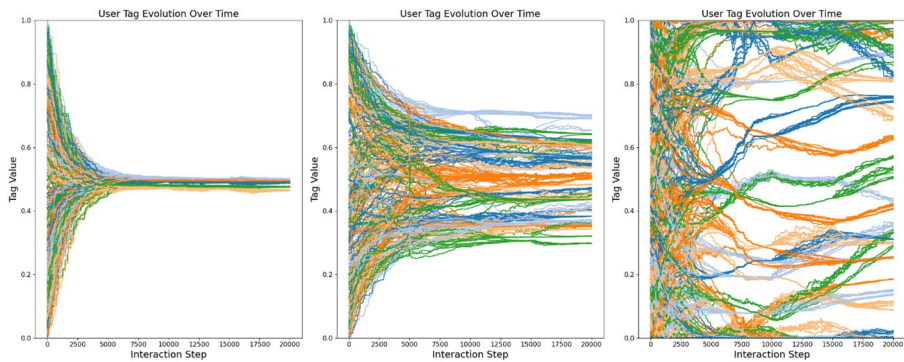


Fig. 4 Evolution of user features over the number of interactions under three different simulated scenarios: random suggestions (leftmost graph), sorting recommendations based on cosine similarity (center graph), cosine similarity with normalization (rightmost graph). The 5 colors in each graph represent the 5 different dimensions of all the users' features

of recommendations needed follows a normal distribution with a high mean and standard deviation such that the left tail is nearly zero for negative recommendations, a kernel analysis indicates that the data better aligns with a Weibull distribution. Nevertheless, the 95% confidence intervals for the average number of recommendations needed, computed by inferring the Weibull distribution via bootstrapping, are very close to the intervals computed with the t-distribution. For example, to reach a 100% sensitivity level, the 95% CI assuming a normal distribution is [21.37; 33.14] while the 95% CI assuming a Weibull distribution is [21.37; 37.89]. The averages are 27.25 and 27.33, respectively. As a final note, running bootstrapping with different seeders affects the resulting CI. Data regarding the confidence intervals computed assuming a Weibull distribution are available upon request.

We conducted convergence analysis through simulations with real data collected during the deployment of the platform to evaluate the convergence properties of the algorithm. The update procedure for the vectors representing users and services is a convex combination of previous values and interaction data, governed by learning and decay rates (see Eqs. 3 and 4). In Fig. 4, we illustrate the evolution of user features over the number of interactions. We simulated three different scenarios, all using the same underlying parameters: 67 users, 150 services, a learning rate of $\lambda = 0.005$, the introduction of new services into the platform in groups of 5 every 5000 interactions, and both user and service vectors being N-dimensional with $N=5$ (hence the 5 colors representing the 5 different dimensions of all the users' features). The choice of 67 users and 150 services comes from real data stored in the platform's database.

The study simulated three recommendation conditions:

1. **Random Suggestions:** In this scenario, continuous updates quickly pushed feature values toward an average (around 0.5). This rapid convergence leads to homogenization of user and service profiles, losing individual identities

and producing uniform recommendation lists where the same service might be repeatedly suggested.

2. Cosine Similarity Sorting: Here, feature values converged within a broader range (approximately 0.3 to 0.7), maintaining some differentiation. However, the tendency to gravitate toward mid-range values can fail to fully capture the true diversity of user preferences. To counter this, normalization was applied, which at specific points (after 5000, 10000, and 15000 interactions) helped some features diverge, indicating a healthier learning process even as the filter bubble effect risk re-emerges.
3. Cosine Similarity with Normalization: With normalization, the full range of values from 0 to 1 is utilized, clearly distinguishing between users and services. This approach better preserves the differences in user preferences (for example, one user might have a “Culture” feature of 0.7 while another has 0.2), reducing the risk of homogenization. Although the filter bubble effect is not entirely eliminated, it is mitigated by the introduction of new services along with the normalization process.

In the simulations showcased here, we aimed to demonstrate the convergence of the features, and the chosen parameters were selected to best illustrate this phenomenon. For instance, when we increased the dimensions to $N=20$, the resulting number of series made it difficult to follow the convergence in the rightmost plot. Second, the number of interactions required to achieve a semblance of convergence depends on the learning and decay rates, higher learning rates accelerate convergence. Additionally, variations between simulations are expected due to the inherent randomness, which may cause the features to converge to different final values. However, it is noteworthy to observe the homogenization process, particularly in how the final converging values differ in range across scenarios. In the first scenario on the left, the values cluster around the average value of 0.5. In the middle scenario, which is before applying normalization, the values range from 0.3 to 0.7. In the rightmost scenario, the values are spread across the full range from 0 to 1.

The differences in these convergence patterns underscore the importance of the recommendation strategy in influencing user experience. While random suggestions can lead to undesirable homogenization, sorting based on cosine similarity, especially when combined with normalization, can preserve and enhance the distinctiveness of user profiles and service attributes, leading to more relevant and personalized recommendations. By analyzing the top- k services shown to users as k varies, we determined that the algorithm achieved an average sensitivity of 79.74% with $k = 20$, meaning that nearly 80% of the users’ choices were within the top 20 recommended items. This demonstrates the effectiveness of the recommender engine in providing relevant suggestions. These findings are crucial for understanding the strengths of our platform and guiding future enhancements to maintain and improve recommendation accuracy.

5.2.1 Comparative evaluation with rating-based baseline

To further assess the effectiveness of our hybrid recommendation strategy involving the introduction of previously unused services, a further experimental evaluation has been conducted involving again the same sample of users of the first test (67 users) and replicating the experimental protocol previously applied to evaluate our hybrid recommender engine. We systematically compared our hybrid recommender algorithm with a traditional rating-based collaborative filtering system, using the Surprise recommendation library as a baseline. Each user received recommendations generated by both the hybrid recommender and the Surprise-based baseline. We evaluated recommendation accuracy using Precision, Recall, and F1-Score metrics.

Precision represented the proportion of recommended services positively endorsed by users, while Recall indicated the fraction of all positively evaluated services successfully recommended. The traditional rating-based recommender demonstrated strong performance, achieving an average F1-score of 0.79, while our hybrid recommender approach yielded a comparably robust average F1-score of 0.77.

Going deeper, the rating-based baseline achieved a slightly higher Precision (0.81) than our hybrid approach (0.78; $\Delta=0.03$) and a marginally higher Recall (0.77 vs. 0.76; $\Delta=0.01$). A paired two-tailed t-test conducted over the 67 individual user profiles revealed that neither of these gaps was statistically significant (Precision: $t=1.12$, $p=0.27$; Recall: $t=0.58$, $p=0.56$). The patterns nevertheless illuminate the underlying trade-off between accuracy and exposure. The baseline optimises purely for rating likelihood and therefore concentrates its recommendations on a narrower subset of highly rated, high-frequency items, which naturally inflates Precision while maintaining respectable Recall among already popular services. Conversely, the hybrid method, by design, injects less-frequent and contextually diverse items into each list. This deliberate diversification produces a modest Precision penalty (3% points) that is not matched by a comparable loss in Recall, demonstrating that exposure to novel items can be increased without materially reducing the system's ability to retrieve relevant services.

This marginal performance gap can be attributed to the baseline's focus on optimizing rating prediction accuracy and its tendency to favor frequently rated (popular) items, which often aligns recommendations with well-established user preferences and thus boosts these accuracy metrics. Nevertheless, this F1-score difference was not statistically significant ($p>0.05$).

Instead, substantial and statistically significant differences emerged in the aspects of recommendation diversity and novelty. Item coverage increased markedly from about 48% in the baseline Surprise-based system to approximately 65% with our hybrid approach. In parallel, novelty assessment, demonstrated that our hybrid method included nearly 25% more novel items compared to the baseline.

Such improvements are particularly crucial for promoting underrepresented services in fragile tourism territories, suggesting that the hybrid's slight trade-off in F1 is justified by its broader benefits in this context.

6 Conclusions

We developed the B4R platform to increase the visibility of under-promoted services that lack the means for effective self-promotion, a concern to which the B4R project is particularly attentive. Additionally, we made deliberate choices, such as foregoing ratings that can sometimes lead to unfair outcomes, such as over-representing popular destinations or under-representing less well-known ones contributing to problems such as “overtourism” and “undertourism” (Banerjee et al. 2023).

Unlike conventional tourism platforms that primarily aggregate attractions, this platform has been engineered to address specific challenges inherent to fragile territories by integrating functionalities that both adapt to individual user preferences and actively contribute to regional branding. Such hybrid systems have been shown to mitigate popularity biases and provide more equitable exposure to underrepresented items, thus effectively addressing issues common in marginal territories (Abdollahpouri et al. 2019). Moreover, our dedicated “Stories” section enables local stakeholders, businesses, event organizers, and residents, to share authentic narratives, bridging visitors with local identities. This storytelling strategy has proven to be effective in highlighting the uniqueness of lesser-known destinations and fostering a deeper visitor engagement and sense of place attachment (Moscardo 2020).

By tracking every user interaction, even if only a handful of visitors engage with a given service, our recommendation engine increases the chances that small providers appear within top-k lists respect to traditional systems. In doing so, we overcome the “hidden gem” problem that ordinarily afflicts marginal territories. Moreover, although visitor-flow analytics are commonly used in high-traffic contexts, the same techniques applied at smaller scales yield even more precise guidance in under-touristed areas. For instance, observing that a recent group of hikers visited a particular agritourism allows platform operators to prompt neighboring wineries or artisanal workshops at exactly the right moment. In this way, our focus on novelty injection and item coverage does not simply replicate existing tourism marketing strategies but elevates low-density attractions from the few interactions that do occur, providing a strategic advantage in marginal territories that cannot rely on volume alone.

We demonstrated how state-of-the-art features can drive the implementation of a platform and a recommender engine to enhance user experience through personalized suggestions, and how the creation of itineraries can help direct tourists’ attention towards networking opportunities and inland destinations in the test area where the platform was deployed. By employing surveys, a series of simulations, a comparative analysis with a traditional recommender approach, and sensitivity analyses, we evaluated the performance of our recommender engine under various conditions, offering insights into how user and service features evolve over time and how these affect the quality and diversity of recommendations.

The results from the survey show that our platform can improve the planning process. In particular, we were interested the most in the answers to the questions “Do you think that trying our B4R platform might have sparked your interest in visiting the Marche hinterland?” and “Does the B4R platform enhance your tourist experience in the area?” and the responses were satisfactory, albeit we have a small sample size. The sensitivity analysis demonstrated that our recommender engine is highly

effective in placing relevant suggestions within the top-k recommendations. The results showed that nearly 80% of user choices were captured within the top 20 items recommended, underscoring the system's ability to prioritize items that closely align with user preferences. Moreover, the sensitivity analysis highlighted the critical role of recommendation list length in balancing relevance and diversity. By carefully tuning the number of recommendations, it is possible to achieve high sensitivity while avoiding the pitfalls of overwhelming users with too many options. It is unlikely that users chose the top results simply because they did not want to scroll further, as some selected services positioned as far down as the 90th and 92nd slots. The broad range of choices observed suggests that our results are not merely the outcome of chance, however, conducting similar analyses on different datasets or in different application domains may help validate the generalizability of our findings.

Moreover, through simulations, we explored the convergence of user and service features under different recommendation strategies. The findings revealed that random suggestions lead to a rapid homogenization of feature values around the mean, cosine similarity-based recommendations introduced more variation in feature values but still showed a tendency toward mid-range convergence, which may not fully reflect the diversity of user preferences and may lead to filter bubble effect, and correcting the features via normalization successfully expanded the range of feature values, allowing for greater differentiation between users and services. The introduction of new services into the platform disrupts the convergence process and helps counteract the filter bubble effect. This approach mitigated the risk of homogenization and preserved a more diverse set of recommendations.

Finally, to validate the performance of our hybrid recommender algorithm, we conducted a comparative analysis using a traditional rating-based recommender engine as a baseline.

The baseline system achieved an F1-Score of 0.79, while our hybrid approach maintained a comparably robust average F1-score of 0.77. Additionally, our analysis revealed that our hybrid method significantly increased item coverage, from about 48% in the baseline to approximately 65% ($p < 0.01$) and incorporated 25% more novel items into the recommendation lists. These enhancements translate into greater user engagement and a more personalized travel planning experience, ensuring lesser-known attractions and services remain visible, even when user-generated data (e.g., reviews, ratings) is scarce, which are central to our objectives.

6.1 Key learnings from the user study

The empirical investigation provided valuable insights into the effectiveness and practical implications of our platform's hybrid recommender system. The data from our user survey underscore that the B4R platform significantly increased participants' interest in exploring the Marche hinterland, highlighting the system's ability not merely to reflect existing tourist interest but actively to stimulate new demand. Nearly three-quarters of respondents indicated enhanced curiosity and a greater likelihood of visiting previously under-promoted locations, validating the potential of recommender systems to influence visitor behaviors positively in marginal territories. Additionally, the performance metrics showed that our hybrid approach suc-

cessfully balanced relevance and diversity. This suggests that integrating less-rated services within recommendation lists can significantly boost visibility for small or lesser-known providers without compromising the quality of recommendations. Lastly, the simulation analyses revealed critical dynamics concerning recommendation strategies. Purely similarity-based approaches tended towards homogenization of suggested features, risking the formation of filter bubbles, while periodic normalization and controlled introduction of new items effectively maintained a broad range of distinct features, preserving diversity and avoiding convergence issues. These results suggest critical avenues for future research, emphasizing the need for explicit strategies such as curiosity metrics and normalization heuristics in recommender design, particularly for contexts characterized by limited data availability and fragile socio-economic conditions.

6.2 Limitations and future work

The findings from this research have significant implications for the design and deployment of recommender systems in various contexts. The importance of maintaining diversity in recommendations cannot be overstated, particularly in domains where user preferences are highly varied and subject to change. However, each use case must be studied with care to understand whether a normalization process such as ours could be beneficial in other recommender systems, particularly those prone to filter bubbles. While this research provides valuable insights, several limitations should be acknowledged. The simulations were conducted with specific parameter settings, and while they illustrate key concepts, the results may vary with different configurations. Additionally, the assumption that we have access to the complete set of user preferences is a simplification. In real-world applications, user preferences may be incomplete or evolve in ways that are not fully captured by the system, particularly if we fix the number of recommendations shown to users. Future work could explore adaptive algorithms that dynamically update and refine user profiles based on new information. Another avenue for future research is the exploration of hybrid recommender approaches that combine the strengths of cosine similarity and normalization with other techniques, such as other collaborative filtering techniques or deep learning models, to further enhance recommendation accuracy and diversity.

Another key limitation of our current evaluation regards the fact that the assessment was based on a comprehensive questionnaire that primarily measured the overall usefulness and user experience of the integrated platform. This method, while effective in capturing the holistic impact of the system, does not isolate the contributions of individual, novel features.

Although the integrated design of the B4R platform reflects the synergistic interaction of its components, future work should consider employing more granular evaluation methods, such as controlled experiments or A/B testing, to isolate and quantify the impact of each innovative functionality. For example, an A/B test could be conducted by randomly assigning live users to one of two platform versions: the control version would present itineraries ranked purely by similarity, while the experimental version would inject a curated selection of under-rated or previously unused services into the recommendation list. By tracking metrics such as click-

through rates, the number of points of interest added to itineraries, and overall session duration, researchers could determine whether the inclusion of novel items directly improves user engagement. Such targeted studies would yield quantitative evidence on the performance of individual features and inform subsequent refinements to further enhance recommendation accuracy and user satisfaction.

To further strengthen the proposed insights, we also plan to expand our data collection and conduct more in-depth, longitudinal analyses that will involve tracking visitor flows, through integrated GPS functionalities, over time to determine whether the platform effectively channels new tourists toward peripheral destinations by comparing historical visitor data (where available) with post-implementation trends. In addition, we will administer follow-up surveys to assess how users perceive the authenticity and distinctiveness of these destinations and to investigate whether their decision to visit unconventional places is influenced by the platform's recommendations.

Two key takeaways from this research are the significant ineffectiveness of random recommender engines, and the underlying factors identified in the convergence analysis of the recommender engine, that provide crucial insights into the systems' performance, emphasizing the importance of understanding and exploiting these factors to enhance the effectiveness and user satisfaction of recommender platforms. As the digital landscape continues to grow and diversify, we believe the principles and techniques explored in this study will remain relevant to the development of next-generation recommender systems.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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